Zakarya Boudraf

Junior Machine Learning Engineer

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PROFILE

Machine-learning graduate student completing an M.Sc. in Internet of Things at the University of Salerno (graduating 24/07/2026), with a prior M.Sc. in Data Engineering and Web Technologies earned Magna Cum Laude (ranked 2nd of 72). Research experience spans audio-visual speech recognition with parameter-efficient fine-tuning of transformer backbones, computer vision, and biomedical signal classification, with hands-on model deployment from microcontrollers to the web. Available from July 2026 for entry-level machine learning roles in Italy (Salerno area, hybrid or remote).

EDUCATION

M.Sc. Internet of Things

University of Salerno, Italy [10/2023 - 07/2026 (expected)] EQF 7

Thesis: efficient visual speech recognition via parameter-efficient fine-tuning (PEFT) of a frozen self-supervised AV-HuBERT backbone.

M.Sc. Data Engineering and Web Technologies

Ferhat Abbas Setif 1 University, Algeria [09/2021 - 07/2023] EQF 7

Magna Cum Laude - ranked 2nd of 72 students. Thesis: self-supervised learning for epileptic seizure detection.

B.Sc. Computing Systems

Ferhat Abbas Setif 1 University, Algeria [09/2018 - 09/2021] EQF 6

Magna Cum Laude - ranked 2nd of 334 students.

RESEARCH AND MACHINE LEARNING PROJECTS

Efficient visual speech recognition via PEFT (M.Sc. thesis, University of Salerno)

[12/2025 - present]

- Adapting a frozen AV-HuBERT transformer backbone to visual-only lip reading using LoRA and adapter modules (PyTorch, fairseq), evaluated on the multilingual MuAViC benchmark.
- Comparing parameter-efficient configurations against full fine-tuning to enable lightweight training of large speech models.

AI-generated art detection · live demo on Hugging Face Spaces

[2024]

- Trained a ResNet CNN (TensorFlow/Keras) on the public AI-ArtBench dataset to distinguish AI-generated from human-made artwork, reaching 94% accuracy.
- Deployed a public interactive demo on Hugging Face Spaces (Gradio) with real-time predictions and confidence scores; presented the project at a University of Salerno research seminar (07/2024).

Self-supervised learning for epileptic seizure detection (M.Sc. thesis, UFAS)

[2022 - 2023]

- Applied self-supervised pretraining to EEG signal classification (TensorFlow/Keras), achieving 99.08% accuracy, 83.33% sensitivity and a 0.06 false-positive rate.

Real-time IoT intrusion detection on STM32 · code

- Built a real-time network intrusion-detection system with an embedded neural network running on an STM32 Nucleo-F401RE microcontroller (C/C++, TinyML).

Deep reinforcement learning for emergency traffic control · code

- Trained deep RL agents to optimize traffic signals across multi-intersection networks with emergency-vehicle prioritization (Python).

Data-centric ML for industrial predictive maintenance

- Systematic literature review of data-centric machine-learning approaches for predictive maintenance on industrial sensor data; presented at the International School of IoT.

Additional projects: IoT fire-detection system with MQTT monitoring and automated fan control (Arduino); commercial website for an architecture studio (2019).

WORK EXPERIENCE

Web developer

Sacom Integrale Pub, Setif, Algeria [06/2022 - 09/2022]

- Delivered client websites end-to-end, independently managing complete projects.

- Gathered requirements and negotiated project details directly with customers.

TECHNICAL SKILLS

Machine learning and deep learning: TensorFlow · Keras · PyTorch · fairseq · CNNs · transformers · self-supervised learning · PEFT (LoRA, adapters) · reinforcement learning

Programming and data: Python · NumPy · Pandas · SQL · C/C++ · Java · JavaScript

Deployment and tools: Git and GitHub · GitHub Actions · Gradio · Hugging Face Spaces

Embedded and IoT: STM32 · Arduino · Raspberry Pi · MQTT · sensor integration

Web: HTML/CSS · React · WordPress

LANGUAGES

Arabic (native) | English | Italian | French